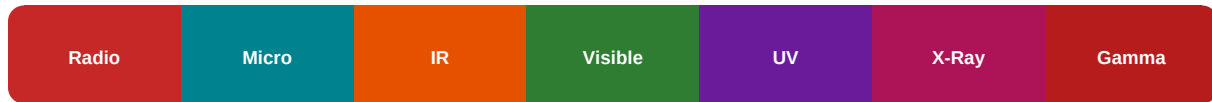


ELECTROMAGNETIC SPECTRUM

Visible Light & Beyond



Subject:	Science
Grade:	Grade 4
Topic:	The Electromagnetic Spectrum
Duration:	45–60 Minutes
Prepared by:	Planrs Academy

■ Learning Objectives

By the end of this lesson, students will be able to:

1. Identify the 7 types of electromagnetic waves in order.
2. Explain the difference between visible and invisible EM waves.
3. Give real-life examples of each wave type from daily Indian life.
4. Recall the ROYGBIV colours of visible light using mnemonics.
5. Understand how wavelength relates to energy across the spectrum.

SECTION 1

Introduction — The World of Waves

What Is the Electromagnetic Spectrum?

Look around you right now. Your Wi-Fi router is sending invisible signals through the air. The TV remote uses invisible heat rays to change channels. When you go to a doctor with a broken bone, X-ray machines see through your skin. After monsoon rain, a beautiful rainbow arches across the sky. All of these — Wi-Fi, TV remotes, X-rays, and rainbows — belong to one enormous family of waves called the **Electromagnetic Spectrum**.

The electromagnetic spectrum is a continuous range of waves that travel at the speed of light (3,00,000 km per second!). These waves carry energy from one place to another without needing any material to travel through — they can even travel through empty space!

■ Key Concept

An electromagnetic wave is made of oscillating electric and magnetic fields that travel through space. Unlike sound waves, they do NOT need air or water to travel.

■ Indian Connection

Think of the Thali (■■■■■) you eat from — it has 7 different dishes. Similarly, the electromagnetic spectrum is like a Thali with 7 types of waves, each with different energy and uses.

The 7 Members of the EM Family



From the longest, gentlest waves to the shortest, most powerful ones, the EM spectrum has seven members. As wavelength *decreases*, energy *increases*.

Wave Type	Wavelength	Energy Level	One Key Use
Radio Waves	km to mm	Lowest	Akashvani radio broadcasts
Microwaves	cm to mm	Low	Microwave ovens, ISRO satellites
Infrared	1 mm – 700 nm	Medium	TV remotes, COVID thermometers
Visible Light	700 – 400 nm	Medium	Everything our eyes see!
Ultraviolet (UV)	400 – 10 nm	High	Vitamin D production, sterilisation
X-Rays	10 – 0.01 nm	Very High	Hospital bone scans
Gamma Rays	< 0.01 nm	Highest	Cancer treatment (AIIMS)

SECTION 2

Visible Light — The Rainbow We Can See

What Is Visible Light?

Of all the waves in the electromagnetic spectrum, **visible light is the only one our eyes can detect**. It is a tiny slice of the entire spectrum, but it gives us all the colours we see — from the fiery red of a sunset to the cool violet of twilight.



The Rainbow — India's Monsoon Gift

When sunlight enters a raindrop during the monsoon, it slows down and bends (refracts). Different colours bend by different amounts because they have different wavelengths. Red bends the least; violet bends the most. As the light exits the raindrop it separates into all seven colours — creating the magnificent **Indradhanush** (■■■■■■■■)!

Colour	Hindi Name	Wavelength	Fun Indian Example
Red	■■■	~700 nm	Red dupatta in Rajasthani dance
Orange	■■■■■■■	~620 nm	Saffron in the national flag
Yellow	■■■■■	~580 nm	Marigold garlands at festivals
Green	■■■	~530 nm	Green in the national flag
Blue	■■■■■	~470 nm	Krishna's blue complexion in art
Indigo	■■■	~445 nm	Indigo dye from the Bengal plant
Violet	■■■■■■■	~400 nm	Jacaranda blossoms in Shimla

■ Mnemonic — Remember ROYGBIV

English: *Richard Of York Gave Battle In Vain*

Hindi: *Raju Orange Friend turned Blue because of Violet in Indore*

Practise saying the mnemonic aloud three times — you will never forget it!

SECTION 3

Deep Dive — Each Wave Explored

Radio Waves

Longest wavelength (km to mm) | Lowest energy

Radio waves are the gentle giants of the spectrum. Their long wavelength lets them travel enormous distances and even bend around hills and buildings. That is why you can listen to All India Radio (Akashvani) in a remote village deep in the mountains!

- **All India Radio (Akashvani)** — music and news to every corner of India.
- **Doordarshan** — India's national television network.
- **Wi-Fi routers** (2.4 GHz & 5 GHz) — the internet at home and school.
- **Mobile networks** (Jio, Airtel, BSNL) — connecting all 1.4 billion Indians!
- **ISRO deep-space communication** — talking to spacecraft millions of km away.

■ Remember

Radio waves are completely safe for daily use because of their low energy.

Microwaves

Wavelength: cm to mm | Energy: Low-medium

Microwaves have slightly shorter wavelengths than radio waves, giving them a bit more energy. They can make water molecules vibrate very rapidly — this vibration produces heat, which is exactly how a microwave oven warms up your food. No fire needed; the food heats itself from inside!

- **Microwave ovens** — heat dal makhani in minutes by vibrating water molecules.
- **ISRO satellites** (INSAT, GSAT) — use microwaves to beam TV and phone signals.
- **IMD weather radar** — tracks monsoon clouds and cyclones like Fani and Amphan.
- **GPS navigation** — the Ola/Uber map on your phone uses microwave satellite signals.

■ Remember

Microwave ovens are safe because the door has a metal mesh that blocks waves from escaping.

Infrared (IR)

Wavelength: 1 mm – 700 nm | Energy: Medium (heat energy)

Infrared waves sit just below red light. Our eyes cannot see them, but our skin can feel them as warmth! Every warm object emits infrared radiation. When you stand near a hot tandoor oven or feel the sun on your skin, that gentle heat is infrared at work.

- **TV remotes** — send infrared pulses to the sensor on your television set.
- **COVID forehead thermometers** — measure body temperature without touching skin.
- **Tandoor ovens** — bake naan and roti using intense infrared heat.
- **ISRO satellites** — study Earth's land use and forest cover using infrared cameras.
- **Night-vision cameras** — used by the Indian Army on borders.

■ Remember

You must point a remote directly at the TV because infrared travels in straight lines.

Ultraviolet (UV)

Wavelength: 400 – 10 nm | Energy: High

UV waves have more energy than visible light. They are invisible but powerful. The sun emits UV rays, which is why you get sunburned if you play cricket outside all afternoon. However, gentle morning sunlight triggers your skin to make Vitamin D — vital for strong bones.

- **■ Morning sunlight** — 15–30 min helps produce Vitamin D (no cream needed!).
- **■ Indian currency notes** — have invisible UV security markings to detect fakes.
- **■ Hospital UV lamps** — kill bacteria and viruses in operation theatres.
- **■ Bees see UV patterns** on flowers — like landing strips guiding them to nectar!
- **■ UV sterilisers** — used in water purifiers (like Kent, Aquaguard).

■ Remember

Apply sunscreen (SPF 30+) before afternoon outdoor play to protect against UV damage.

X-Rays

Wavelength: 10 – 0.01 nm | Energy: Very High

X-rays carry so much energy that they can pass right through soft tissues like skin and muscle, but are stopped by dense materials like bone and metal. This makes them perfect for medical imaging — doctors can see inside your body without any surgery! Wilhelm Röntgen discovered X-rays by accident in 1895.

- **■ Hospital bone scans** — check for broken bones and fractures.
- **✈ ■ Airport security scanners** (every Indian airport) — see inside bags.
- **■ ISRO & DRDO** — use X-rays to check rocket engine parts for tiny cracks.
- **■ Dental X-rays** — check teeth roots and cavities painlessly.
- **■ AstroSat (ISRO)** — India's space telescope studies X-rays from distant stars.

■ Remember

Medical X-ray doses are tiny and carefully controlled — they are safe for patients.

Gamma Rays

Shortest wavelength (<0.01 nm) | HIGHEST energy

Gamma rays are the most powerful waves in the entire universe. They can pass through almost anything. Despite their danger, scientists have found life-saving medical uses for them. They are also produced naturally by nuclear reactions in stars and during massive stellar explosions called supernovae.

- **■ AIIMS & Tata Memorial Hospital** — 'Gamma Knife' destroys cancer tumours.
- **■ BARC Trombay, Mumbai** — India's nuclear research centre studies gamma radiation.
- **■ ISRO AstroSat** — detects gamma-ray bursts from colliding neutron stars.
- **■ Food irradiation** — gamma rays sterilise spices and onions for export.
- **■ Medical equipment sterilisation** — makes surgical tools germ-free.

■ Remember

Gamma rays require thick lead or concrete shielding — always follow safety procedures.

SECTION 4

How Does It Work? — Three Real-Life Explanations

1. How Does a TV Remote Work?

When you press a button on your TV remote, a tiny LED inside flashes very quickly with infrared light — too fast and too invisible for your eye to notice. These IR pulses travel in a straight line at the speed of light (3,00,000 km/s!) and hit a small sensor on the front of your television set. The TV's computer reads the pattern of flashes and converts it into a command — change channel, increase volume, switch input, and so on.

■ Key Insight

This is why you MUST point the remote directly at the TV — infrared travels in straight lines and cannot bend around corners (unlike radio waves, which can). If something blocks the path (your hand, a pillow), the signal is interrupted!

2. Why Does Mummy Say 'No Afternoon Sun'?

The sun produces UV rays in three types — UVA, UVB, and UVC. Earth's ozone layer absorbs most UVC and some UVB. In the afternoon, the sun is directly overhead, so the UV rays travel a shorter path through the atmosphere. More UV reaches the surface, and it is stronger! Prolonged exposure causes sunburn (skin turns red and sore) and over many years can damage skin cells.

■ Morning Sun (Good!)

Angle: Low → more atmosphere to pass through → weaker UV
UVB helps your skin make Vitamin D — essential for health
Best time: 7–9 AM for 15–30 minutes.

■ Afternoon Sun (Careful!)

Angle: High → shorter path → stronger UV.
Causes sunburn within 15–20 min for lighter skin tones.
Wear sunscreen SPF 30+ and a hat if going out.

3. How Does an X-Ray See Your Bones?

An X-ray machine fires a beam of X-ray photons through your body onto a special detector on the other side. As the beam passes through, different tissues absorb different amounts:

Tissue / Material	X-Ray Absorption	Appears On Image
Bone (calcium-rich)	High — stops most X-rays	Bright white
Muscle / fat	Medium	Grey shades
Lungs (air-filled)	Very low	Dark / black
Metal (implant, coin)	Very high	Bright white outline

SECTION 5

Classroom Activities & Games

Activity 1 — EM Wave Sorting Game

Look at each item below. Write the correct EM wave type next to it. Choose from: Radio, Microwave, Infrared, Visible Light, Ultraviolet, X-Ray, Gamma Ray.

#	Item / Scenario	EM Wave Type
1	Akashvani All India Radio broadcast	_____
2	TV remote changing channels	_____
3	Monsoon rainbow in the sky	_____
4	Doctor checking a broken arm	_____
5	Sunburn after playing cricket	_____
6	Heating dal in a microwave oven	_____
7	AIIMS Gamma Knife cancer treatment	_____
8	Wi-Fi signal from the router	_____
9	COVID forehead thermometer	_____
10	Bees seeing flower patterns	_____

Answer Key: 1-Radio, 2-Infrared, 3-Visible, 4-X-Ray, 5-UV, 6-Microwave, 7-Gamma, 8-Radio, 9-Infrared, 10-UV

Activity 2 — EM Spectrum Chant

Stand up and say this chant with hand gestures — start with arms spread wide (Radio = long) and bring hands closer together with each wave type (Gamma = fingertips touching):

■ The EM Chant

RADIO → MICRO → INFRA → VISIBLE!

ULTRA → X-RAY → GAMMA!

Say it loud, say it proud — faster each time!

Clap on VISIBLE (the middle) — it is the only one we can see!

Activity 3 — Wave Detective Homework

Go home tonight and find **five objects** that use invisible EM waves. Write each one in the table below, name the wave type, and explain how it works.

Object Found at Home	EM Wave Type	How Does It Use This Wave?
1.		
2.		
3.		

4.		
5.		

SECTION 6

Rapid-Fire Quiz — Test Yourself!

Answer these questions as fast as you can. Answers are provided below.

Q1. Which EM wave does a TV remote use?

■ Infrared waves — they are invisible heat rays that carry the signal.

Q2. Which wave lets doctors see broken bones without surgery?

■ X-Rays — they pass through soft tissue but are blocked by dense bone.

Q3. What causes sunburn after playing cricket in the afternoon sun?

■ Ultraviolet (UV) rays from the sun carry enough energy to damage skin cells.

Q4. Which wave in the EM spectrum has the MOST energy?

■ Gamma Rays — they have the shortest wavelength and highest frequency.

Q5. Name the 7 colours of visible light in order from longest to shortest wavelength.

■ Red, Orange, Yellow, Green, Blue, Indigo, Violet (ROYGBIV).

Q6. Why can you listen to Akashvani radio in remote hilly villages?

■ Radio waves have long wavelengths that bend around hills and mountains.

Q7. What does ISRO's AstroSat telescope detect from space?

■ X-rays and Gamma Rays from distant stars, supernovae, and black holes.

Q8. Which part of the EM spectrum are human eyes able to detect?

■ Visible light only — the small rainbow-coloured band between infrared and UV.

SECTION 7

Summary & Key Vocabulary

Today's Big Ideas

- The EM spectrum is a family of 7 waves that all travel at the speed of light.
- Visible light is the ONLY member of the spectrum we can see with our eyes.
- As wavelength decreases, energy increases — Radio waves are lowest; Gamma highest.
- Radio, Microwave, IR, UV, X-Rays, and Gamma Rays are used every day in Indian life.
- The rainbow (Indradhanush) shows the 7 colours of visible light: ROYGBIV.
- Invisible waves work silently around us — Wi-Fi, remotes, phone signals, and more.

Key Vocabulary

Term	Definition
Electromagnetic Spectrum	The full range of EM waves arranged by wavelength and energy.
Wavelength	The distance between two wave crests; longer = less energy.
Frequency	How many waves pass a point per second; higher = more energy.
Visible Light	The small portion of the EM spectrum human eyes can detect.
ROYGBIV	The seven colours of visible light: Red, Orange, Yellow, Green, Blue, Indigo, Violet.
Infrared	Invisible heat waves below red light; used in TV remotes.
Ultraviolet (UV)	High-energy invisible waves above violet; cause sunburn but also produce Vitamin D.
X-Ray	Very high-energy waves that pass through soft tissue; used in medical imaging.
Gamma Ray	Highest energy EM wave; used in cancer treatment and space telescopes.
Refraction	Bending of light when it passes from one medium to another (e.g., through a raindrop).
Prism	A transparent shape that separates white light into its rainbow colours.

■ Keep Exploring!

The electromagnetic spectrum is one of science's greatest discoveries. Every time you use Wi-Fi, watch TV, warm food in a microwave, or see a rainbow, you are experiencing the power of EM waves! Stay curious, keep asking questions, and remember — the invisible world around you is full of amazing science waiting to be discovered.

Next Steps: Try making a ROYGBIV colour wheel, build a simple sundial to explore light angles, or visit ISRO's virtual museum online to see real satellite instruments!